

Standing Up an Amazon Aurora Database with an Automatically Rotated Password Using AWS Secrets Manager

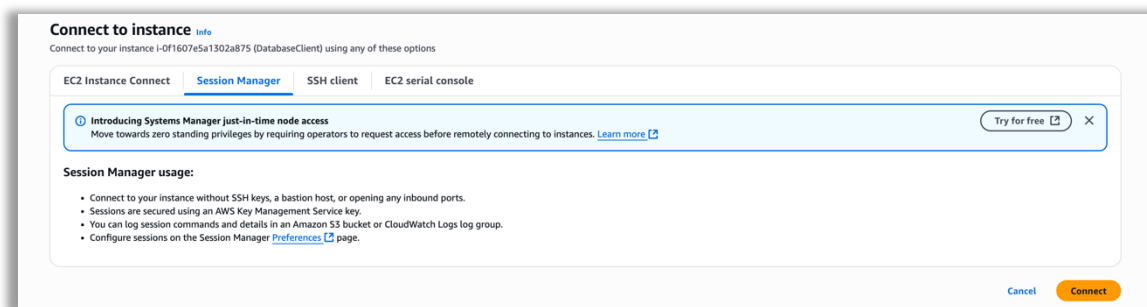
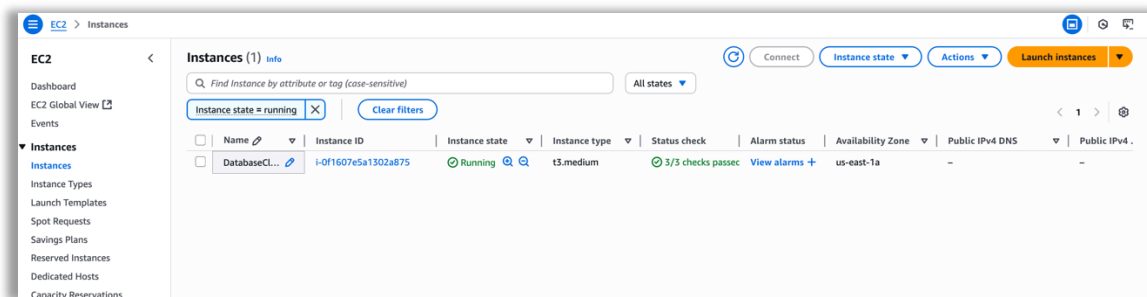
Introduction

This project involved deploying a MySQL-compatible Amazon Aurora database and securing it with automatic password rotation using AWS Secrets Manager. The main objective was to understand how to securely manage and rotate database credentials and to validate successful connectivity and data manipulation through an EC2 instance.

Solution Implementation

Step 1: Verifying MySQL Client Installation

- Navigated to the EC2 Console and connected to the pre-provisioned DatabaseClient instance via Session Manager.

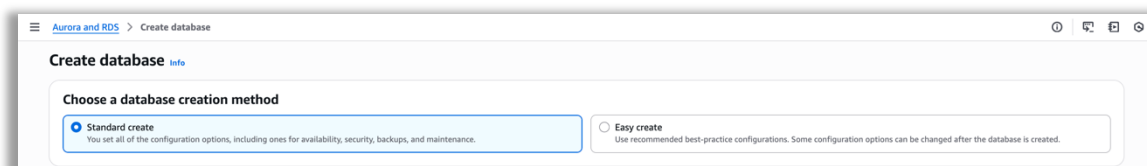


- Verified the installation of the MySQL client using:

```
sh-4.2$ sudo yum list installed | grep mariadb
mariadb.x86_64      1:5.5.68-1.amzn2.0.1      @amzn2-core
mariadb-libs.x86_64 1:5.5.68-1.amzn2.0.1      installed
sh-4.2$
```

Step 2: Creating a MySQL-Compatible Amazon Aurora Database

- Opened the Amazon RDS console and started the Standard Create workflow.



- Chose Aurora (MySQL Compatible) as the engine type.

Engine options

Engine type [Info](#)

☒ Aurora (MySQL Compatible)

☐ Aurora (PostgreSQL Compatible)

☐ MySQL

☐ PostgreSQL

☐ MariaDB

☐ Oracle

☐ Microsoft SQL Server

☐ IBM Db2

Engine version

Aurora MySQL 3.08.2 (compatible with MySQL 8.0.39) - default for major version 8.0

☐ Enable RDS Extended Support [Info](#)

Amazon RDS Extended Support is a [paid offering](#). By selecting this option, you consent to being charged for this offering if you are running your database major version past the RDS end of standard support date for that version. Check the end of standard support date for your major version in the [Amazon Aurora documentation](#).

- Applied the Dev/Test template and provided the following settings:

Templates

Choose a sample template to meet your use case.

☐ Production

☒ Dev/Test

Use defaults for high availability and fast, consistent performance.

This instance is intended for development use outside of a production environment.

- DB Cluster Identifier: employees-database

Settings

DB cluster identifier [Info](#)

Enter a name for your DB cluster. The name must be unique across all DB clusters owned by your AWS account in the current AWS Region.

employees-database

The DB cluster identifier is case-insensitive, but is stored as all lowercase (as in "mydbcluster"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

▼ Credentials Settings

Master username [Info](#)

Type a login ID for the master user of your DB instance.

admin

1 to 32 alphanumeric characters. The first character must be a letter.

Credentials management

You can use AWS Secrets Manager or manage your master user credentials.

☒ Managed in AWS Secrets Manager - most secure

RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☐ Self managed

Create your own password or have RDS create a password that you manage.

[If you manage the master user credentials in AWS Secrets Manager, additional charges apply. See \[AWS Secrets Manager pricing\]\(#\). Additionally, some RDS features aren't supported. See limitations \[here\]\(#\).](#)

Select the encryption key [Info](#)

You can encrypt using the KMS key that Secrets Manager creates or a customer managed KMS key that you create.

aws/secretsmanager (default)

[Add new key](#)

- Master Username: admin (default)
- Enabled Secrets Manager for credential management.
- Selected the default KMS key: aws/secretsmanager
- Storage Configuration: Aurora Standard
- Instance Configuration: db.t3.medium (burstable class)

Instance configuration

The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class [Info](#)

▼ Hide filters

☒ Include previous generation classes

☐ Serverless v2

☐ Memory optimized classes (includes r classes)

☒ Burstable classes (includes t classes)

db.t3.medium

2 vCPUs 4 GiB RAM Network: Up to 2,085 Mbps

- Chose not to create a Multi-AZ replica
- For Connectivity, selected DatabaseClient EC2 instance
- Created new security group: database-sg
- Initial Database Name: employees_db
- Clicked Create database, waited for the cluster to complete provisioning (~15 minutes).

DB identifier	Status	Role	Engine	Region	Size	Recommendations
employees-database	Creating	Regional cluster	Aurora My...	us-east-1	1 instance	
employees-database-instance-1	Creating	Reader instance	Aurora My...	us-east-1a	db.t3.medium	

Step 3: Connecting to the Aurora Database and Importing Data

- Opened the AWS Secrets Manager and retrieved the secret created for the Aurora database.

Secret name	Description	Last retrieved (UTC)
rdscluster-310eabbe-e29b-4255-8c68-82cd07911887	The secret associated with the primary RDS DB cluster: arn:aws:rds:us-east-1:346212964682:cluster:employees-database	-

- Verified secret values (username and password) in JSON format.

Key/value	Plaintext
Secret key	Secret value
username	admin
password	X<F[!mq4[oIQJ.e4YuYNSC_1qdzn

- Returned to EC2 Session Manager and created a dataset:

```

Session ID: cloud_user-xdvcbegh82xr477d7j7l Instance ID: i-0f1607e5a1302a875 Terminate

sh-4.2$ cd
sh-4.2$ ls
employees.txt
sh-4.2$ cat employees.txt
1,Smith,John,john.smith@email.com
2,Doe,Jane,jane.doe@email.com
3,Brown,Sam,sam.brown@email.com
4,Johnson,Pat,pat.johnson@email.com
5,White,Terry,terry.white@email.com
6,Taylor,Alex,alex.taylor@email.com
7,Moore,Linda,linda.moore@email.com
8,Jackson,Ricky,ricky.jackson@email.com
9,Lee,Wendy,wendy.lee@email.com
10,Garcia,Carlos,carlos.garcia@email.com
11,Miller,Zack,zack.miller@email.com
12,Lewis,Lucy,lucy.lewis@email.com
13,Martinez,Miguel,miguel.martinez@email.com
14,Robinson,Neil,neil.robinson@email.com
15,Clark,Ruby,ruby.clark@email.com
16,Rodriguez,Jose,jose.rodriguez@email.com
17,Lewis,Grace,grace.lewis@email.com
18,Lee,Brian,brian.lee@email.com
19,Walker,Paul,paul.walker@email.com
20,Hall,Ema,ema.hall@email.com
21,Allen,Gary,gary.allen@email.com
22,Young,Donna,donna.young@email.com
23,Hernandez,Joan,joan.hernandez@email.com
24,King,Larry,larry.king@email.com
25,Wright,Laura,laura.wright@email.com
26,Lopez,Charles,charles.lopez@email.com
27,Hill,Jessica,jessica.hill@email.com
28,Scott,Earl,earl.scott@email.com
29,Green,Andrea,andrea.green@email.com
30,Adams,Shirley,shirley.adams@email.com
sh-4.2$

```

```

sh-4.2$
sh-4.2$ mysql --host=employees-database-instance-1.cd7jivk5f72.us-east-1.rds.amazonaws.com --user=admin --password employees_db
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 91
Server version: 8.0.39 Sbc99e28

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [employees_db]>

```

```

MySQL [employees_db]> CREATE TABLE Employees
-> (
->     EmployeeId int,
->     LastName   varchar(255),
->     FirstName  varchar(255),
->     Email      varchar(255)
-> );
Query OK, 0 rows affected (0.02 sec)

MySQL [employees_db]> SHOW TABLES;
+-----+
| Tables_in_employees_db |
+-----+
| Employees               |
+-----+
1 row in set (0.00 sec)

MySQL [employees_db]>

```

```

MySQL [employees_db]> CREATE TABLE Employees
-> (
->     EmployeeId int,
->     LastName   varchar(255),
->     FirstName  varchar(255),
->     Email      varchar(255)
-> );
Query OK, 0 rows affected (0.02 sec)

MySQL [employees_db]> SHOW TABLES;
+-----+
| Tables_in_employees_db |
+-----+
| Employees               |
+-----+
1 row in set (0.00 sec)

MySQL [employees_db]>

```

```

MySQL [employees_db]> LOAD DATA LOCAL INFILE '/home/ssm-user/employees.txt' INTO TABLE Employees
->     FIELDS TERMINATED BY ','
->     LINES TERMINATED BY '\n';
Query OK, 30 rows affected (0.01 sec)
Records: 30 Deleted: 0 Skipped: 0 Warnings: 0

MySQL [employees_db]>

```

```

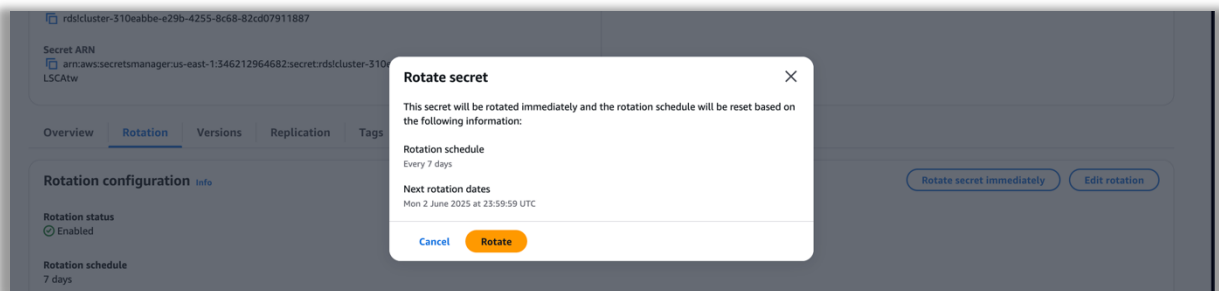
MySQL [employees_db]> LOAD DATA LOCAL INFILE '/home/ssm-user/employees.txt' INTO TABLE Employees
-> FIELDS TERMINATED BY ','
-> LINES TERMINATED BY '\n';
Query OK, 30 rows affected (0.01 sec)
Records: 30 Deleted: 0 Skipped: 0 Warnings: 0

MySQL [employees_db]> SELECT *
-> FROM Employees;
+-----+-----+-----+-----+
| EmployeeId | LastName | FirstName | Email |
+-----+-----+-----+-----+
| 1 | Smith | John | john.smith@email.com |
| 2 | Doe | Jane | jane.doe@email.com |
| 3 | Brown | Sam | sam.brown@email.com |
| 4 | Johnson | Pat | pat.johnson@email.com |
| 5 | White | Terry | terry.white@email.com |
| 6 | Taylor | Alex | alex.taylor@email.com |
| 7 | Moore | Linda | linda.moore@email.com |
| 8 | Jackson | Ricky | ricky.jackson@email.com |
| 9 | Lee | Wendy | wendy.lee@email.com |
| 10 | Garcia | Carlos | carlos.garcia@email.com |
| 11 | Miller | Zack | zack.miller@email.com |
| 12 | Davis | Lucy | lucy.davis@email.com |
| 13 | Martinez | Miguel | miguel.martinez@email.com |
| 14 | Robinson | Neil | neil.robinson@email.com |
| 15 | Clark | Ruby | ruby.clark@email.com |
| 16 | Rodriguez | José | jose.rodriguez@email.com |
| 17 | Lewis | Grace | grace.lewis@email.com |
| 18 | Lee | Brian | brian.lee@email.com |
| 19 | Walker | Paul | paul.walker@email.com |
| 20 | Hall | Emma | emma.hall@email.com |
| 21 | Allen | Gary | gary.allen@email.com |
| 22 | Young | Donna | donna.young@email.com |
| 23 | Hernandez | Joan | joan.hernandez@email.com |
| 24 | King | Larry | larry.king@email.com |
| 25 | Wright | Laura | laura.wright@email.com |
| 26 | Lopez | Charles | charles.lopez@email.com |
| 27 | Hill | Jessica | jessica.hill@email.com |
| 28 | Scott | Earl | earl.scott@email.com |
| 29 | Green | Andrea | andrea.green@email.com |
| 30 | Adams | Shirley | shirley.adams@email.com |
+-----+-----+-----+-----+
30 rows in set (0.00 sec)

```

Step 4: Triggering Credential Rotation

- Navigated to AWS Secrets Manager, selected the relevant secret.
- Clicked on Rotate secret immediately under the Rotation tab.

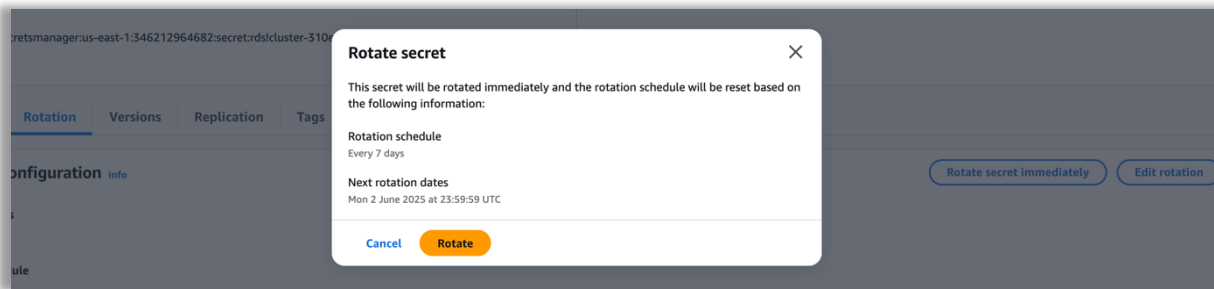


- Waited for the rotation to complete (observed version change from AWSPENDING to AWSCURRENT).
- Tested old credentials from the prior session (as expected, failed).
- Retrieved the new secret value and used the updated password to reconnect:

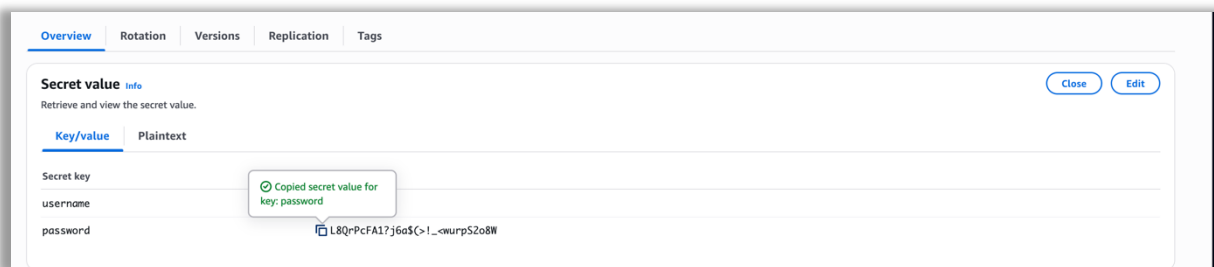
```

sh-4.2$ vim oldpw.txt
sh-4.2$

```



```
sh-4.2$ cat oldpw.txt
fXX<-p$3PZ4XrG4i)Omzf*8[1~c
sh-4.2$
```



```
sh-4.2$ cat oldpw.txt
fXX<-p$3PZ4XrG4i)Omzf*8[1~c
sh-4.2$ mysql --host=employees-database-instance-1.cdwjzivk5f72.us-east-1.rds.amazonaws.com --user=admin --password employees_db
Enter password:
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 212
Server version: 8.0.39 8bc99e28

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [employees_db]>
```

```
sh-4.2$ cat oldpw.txt
fXX<-p$3PZ4XrG4i)Omzf*8[1~c
sh-4.2$ mysql --host=employees-database-instance-1.cdwjzivk5f72.us-east-1.rds.amazonaws.com --user=admin --password employees_db
Enter password:
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 212
Server version: 8.0.39 8bc99e28

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [employees_db]>
```

Conclusion

Through this lab, I successfully:

- Deployed an Amazon Aurora database with managed credentials.
- Leveraged AWS Secrets Manager for secure storage and automatic rotation.
- Established secure EC2-based database access.
- Imported, queried, and verified data from the EC2 instance.

- Triggered and validated password rotation.

This project demonstrates a robust solution for managing database credentials securely using native AWS services, aligning with industry best practices for operational security and automation.
